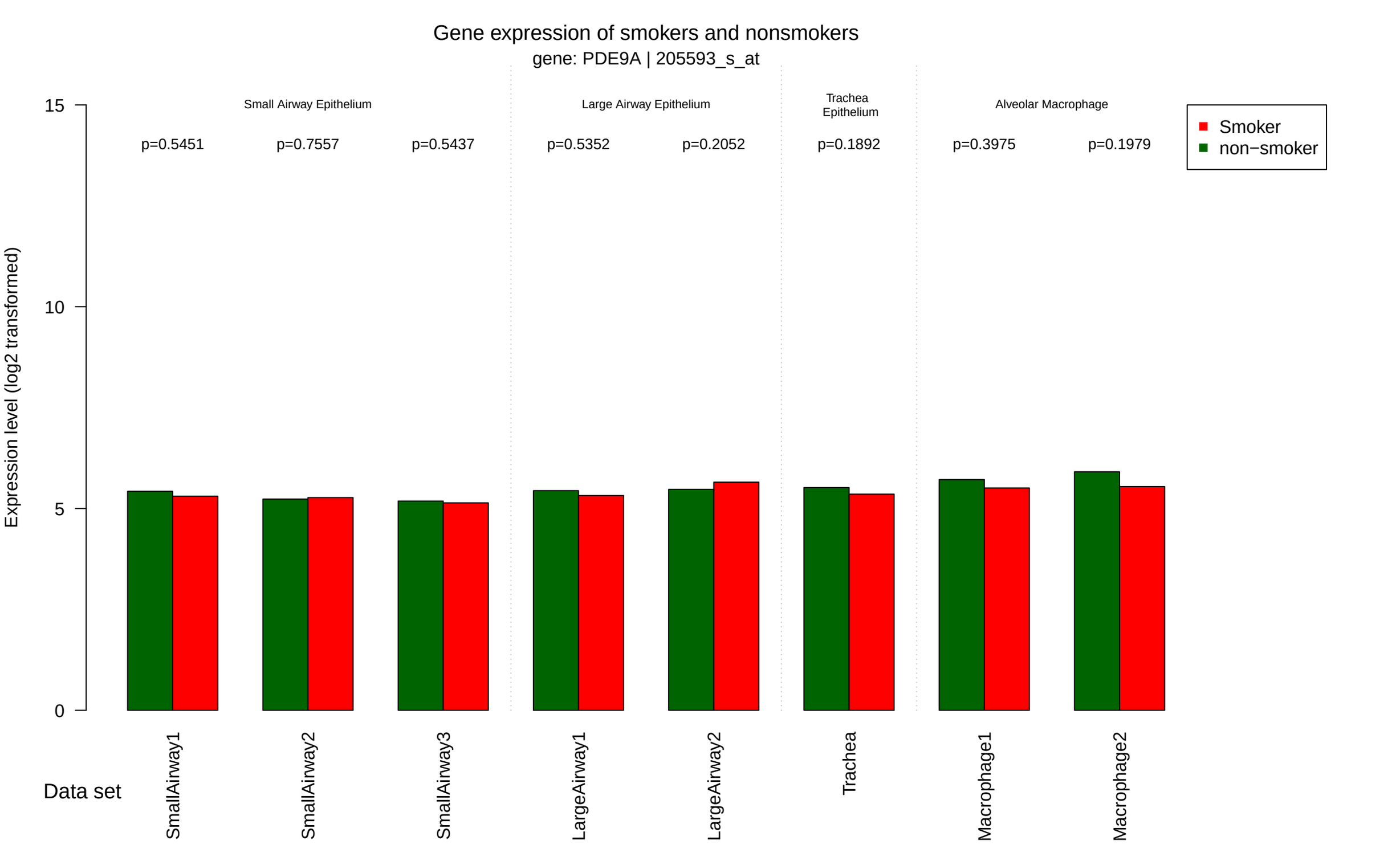
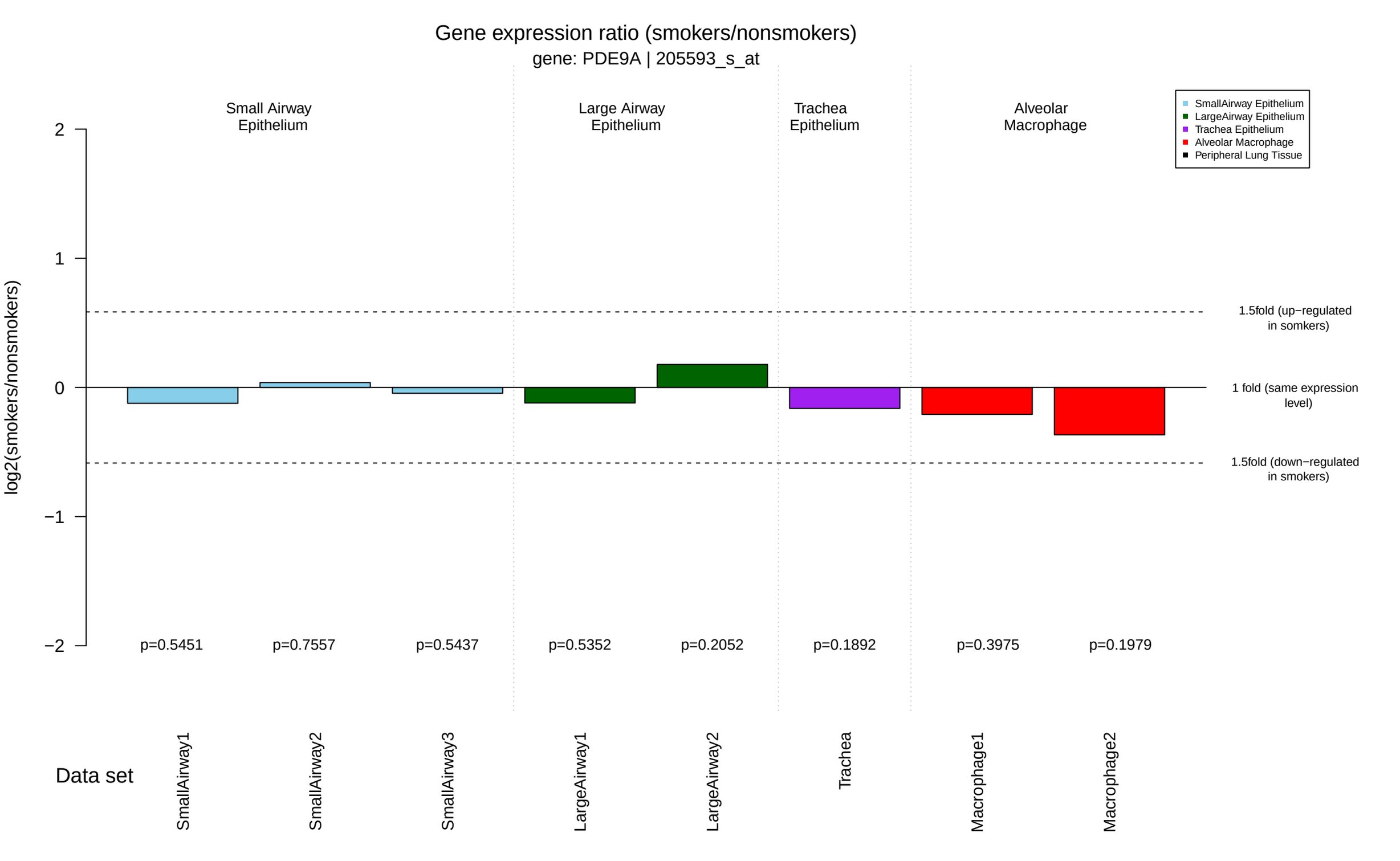
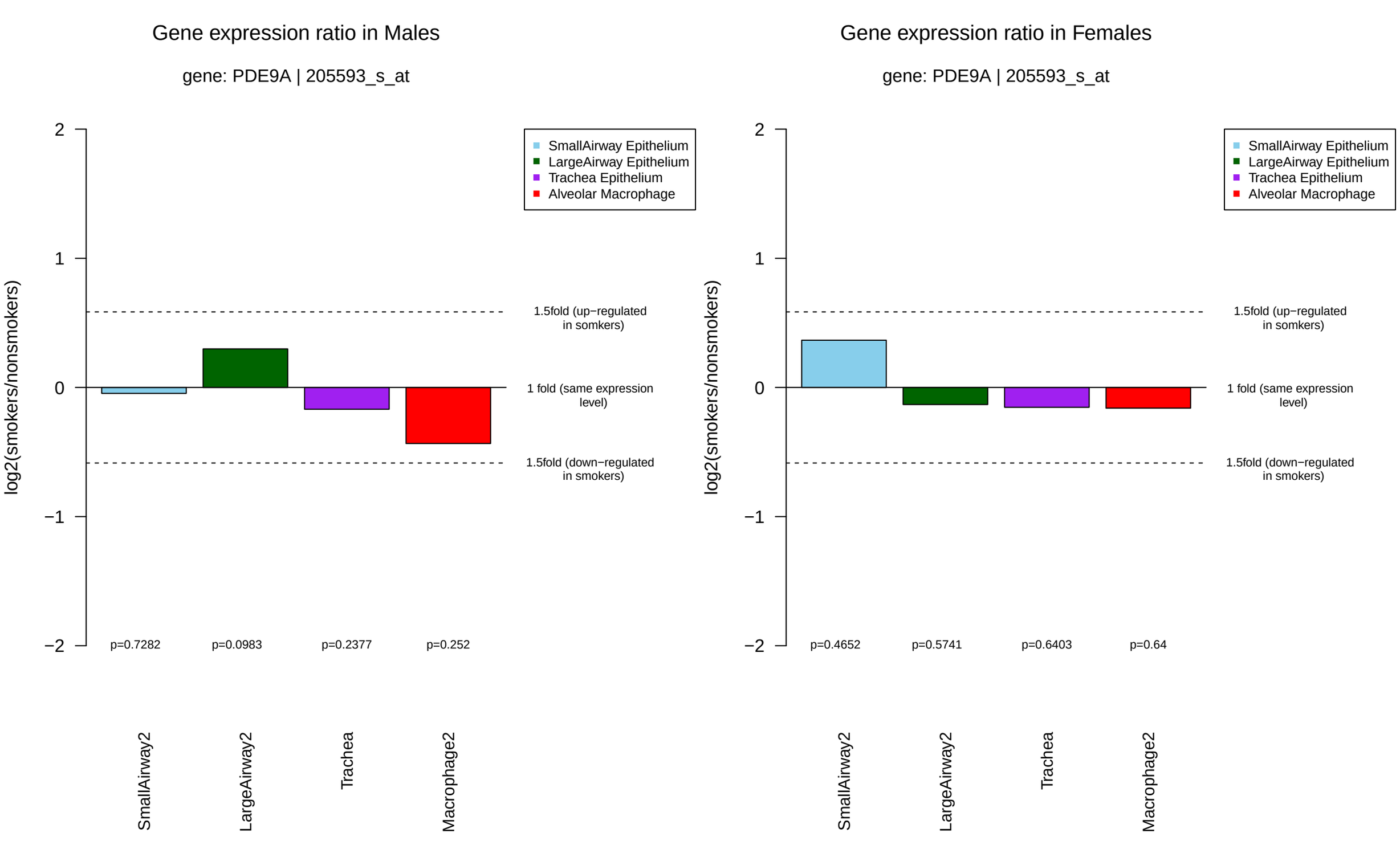
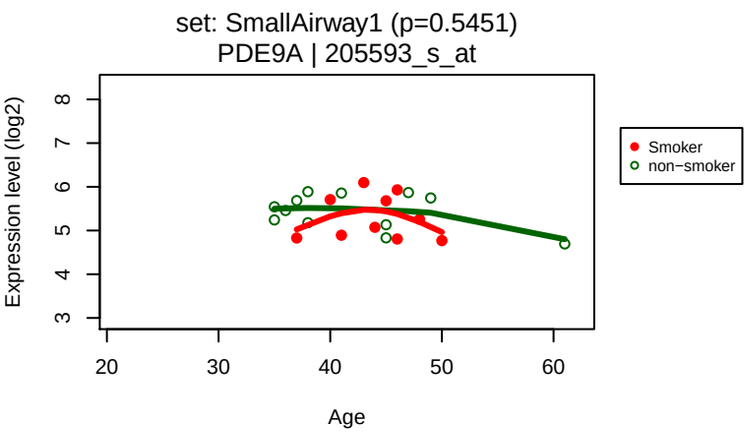
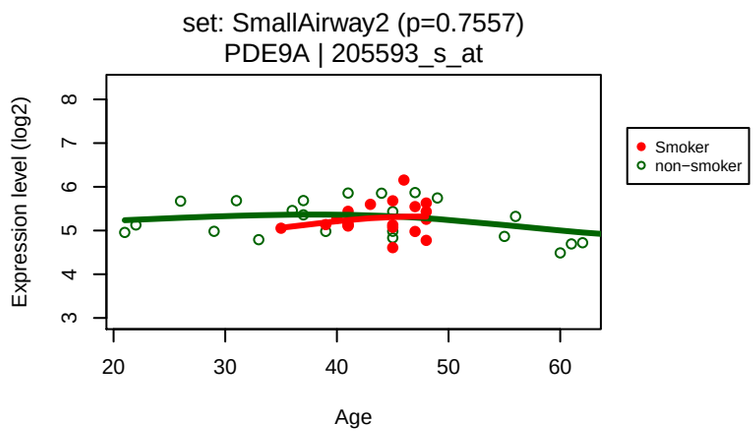
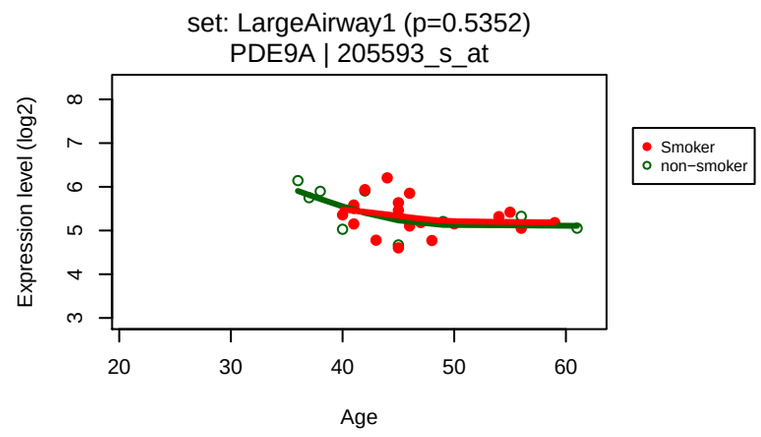

Gene expression of smokers and nonsmokers
gene: PDE9A | 205593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.5451
p=0.7557
p=0.5437
p=0.5352
p=0.2052
p=0.1892
p=0.3975
p=0.1979
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

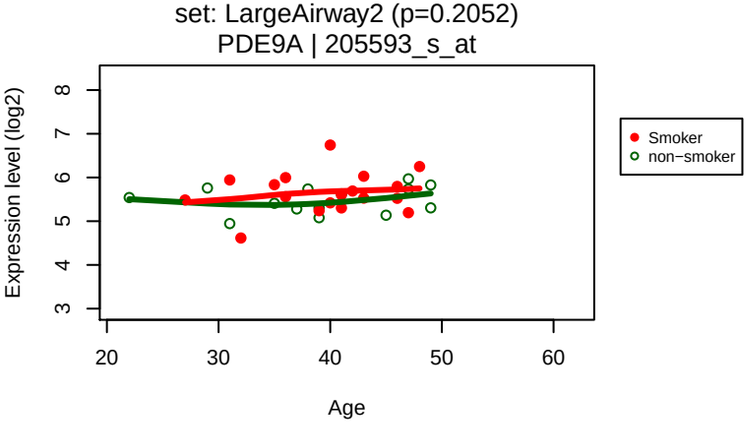

Gene expression ratio (smokers/nonsmokers)
gene: PDE9A | 205593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5451
p=0.7557
p=0.5437
p=0.5352
p=0.2052
p=0.1892
p=0.3975
p=0.1979
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

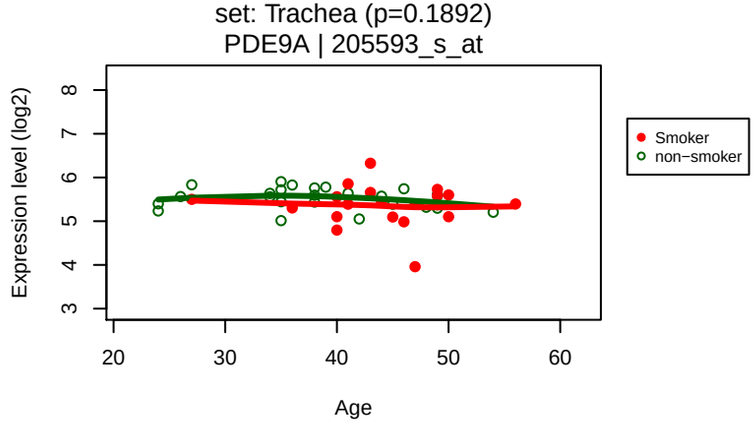

Gene expression ratio in Males
gene: PDE9A | 205593_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7282
p=0.0983
p=0.2377
p=0.252
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE9A | 205593_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4652
p=0.5741
p=0.6403
p=0.64
SmallAirway2
LargeAirway2
Trachea
Macrophage2

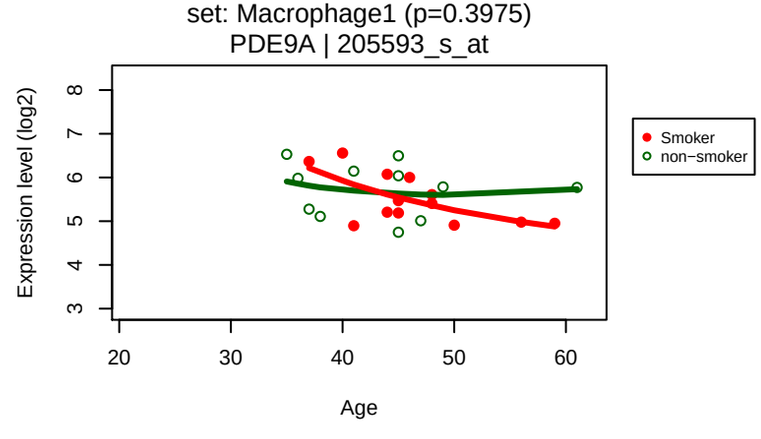

set: SmallAirway1 (p=0.5451)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker

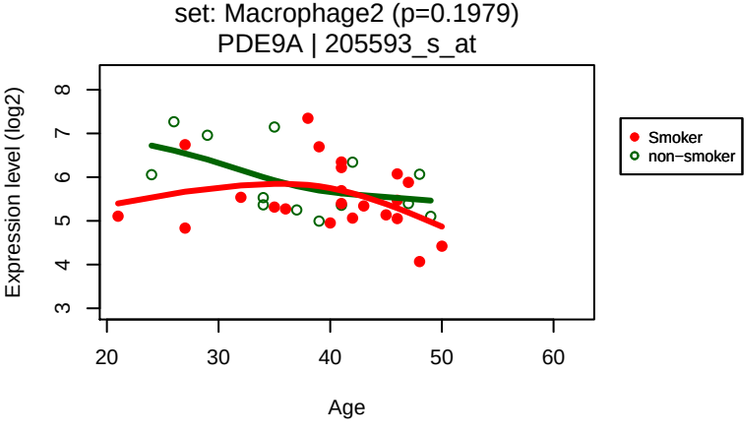

set: SmallAirway2 (p=0.7557)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker

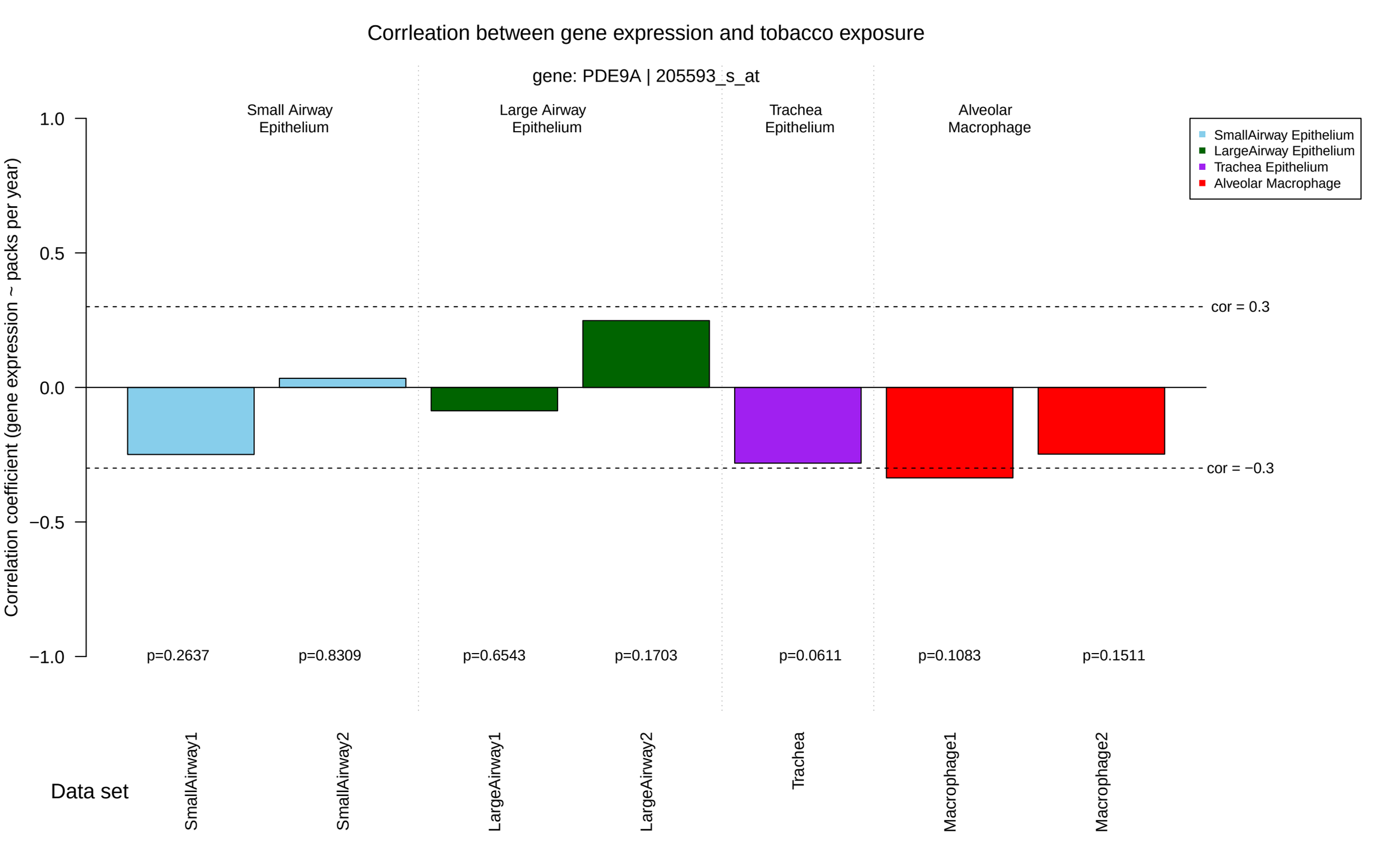

set: LargeAirway1 (p=0.5352)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2052)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1892)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3975)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1979)
PDE9A | 205593_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE9A | 205593_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2637
p=0.8309
p=0.6543
p=0.1703
p=0.0611
p=0.1083
p=0.1511
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2